

Grade 11 Biology Unit Three: Enzymes

3.1. What are enzymes?

At the end of this section, students will be able to:

- Define enzymes and activation energy
- Explain how enzymes work
- Describe the catalysis reaction of enzymes, activities and substrates

3.2 Properties and functions of enzymes

After the successful completion of this section, the student will be able to:

- Identify the properties of enzymes
- Explain the action of each property
- Describe the functions properties

3.3 Protein structures

After the successful completion of this section, the student will be able to:

- Explain the structure of proteins.
- Determine the protein's primary, secondary, and tertiary structures.
- List the levels of protein function

3.4 Enzyme substrate models

After the successful completion of this section, the student will be able to:

- Identify enzyme substrate models
- Explain the active site of enzymes

3.5 Enzyme regulation

After the successful completion of this section, the student will be able to:

- Explain enzyme regulation
- Distinguish activator and inhibitor enzymes
- Describe substrate and bonding

3.6 Types of enzymes

After the successful completion of this section, the student will be able to:

- List types enzymes
- Describe the function of each type of enzyme
- Define activities of each type of enzyme

3.7 Factors affecting enzyme action

After the successful completion of this section, student will be able to:

- List factors affecting enzyme actions
- Describe how each factor affects enzyme action
- Discuss on how to optimize the factors

3.8 Enzyme kinetics

After the successful completion of this section, students will be able to:

- Explain enzyme kinetics
- Describe the steps enzyme kinetic
- How enzyme kinetics works?

3.9 Application of enzymes in industries and their benefits

After the successful completion of this section, the student will be able to:

- Explain applications of enzymes
- List areas of enzyme applications
- Identify and discuss locally applied enzymes

3.10 Malting in Ethiopian tradition

After the successful completion of this section, the student will be able to:

- Explain the what and why of malting for alcohol drinks
- List some of the locally prepared types of malting
- Define the nature and function of malting for alcoholic production